

Gamma Duality, Total Positivity, and Lambert-Localized Moments in the Fabius–Rvachev System

q -binomial multiple-zeta expansions, spectral subordinators, phase-resolved endpoint tilting, and large-order transfer

Frontier research report for Vladimir Reshetnikov’s ProveIt project

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Abstract

The Fabius–Rvachev corpus already contains a remarkably complete triangle of representations: a dyadic random series, exact q -binomial arithmetic, and an infinite product of sinc functions whose negative-Laplace endpoint asymptotic contains a one-periodic fluctuation and the branch W_{-1} of Lambert’s function. This report develops a new layer connecting those three sides. All claims marked “proved here” are supplied with proofs in the present manuscript; they are not claimed to have been formalized in Lean yet.

The first result expands every even moment simultaneously in Gaussian q -binomial coefficients and symmetrized multiple zeta values. The Wick-rotated sinc determinant is then shown to lie in the type-I Laguerre–Pólya class. Consequently its coefficient sequence is Pólya-frequency of infinite order, and every shifted Jensen polynomial is real-rooted; this gives an infinite hierarchy of sharp moment inequalities. The same determinant proves that the Rvachev law is strictly sub-Gaussian with the optimal variance proxy, although its real-axis zeros rule out infinite divisibility.

The principal structural theorem identifies the reciprocal Wick-rotated product as a Laplace transform. Every positive power is the transform of an explicit generalized gamma convolution with discrete Thorin measure $\tau \sum_{m \geq 1} (1 + v_2(m)) \delta_{m^2}$. Its cumulants are $\tau(r-1)! \zeta(2r)/(1-4^{-r})$, it satisfies an exact $q = 1/4$ perpetuity, and the parity of its cumulative Thorin mass is precisely the Thue–Morse sign of the Fourier lobes. The associated Lévy heat kernel admits not only the usual Mellin expansion with log-periodic oscillation, but an exact Jacobi-theta completion whose error begins at $-2\sqrt{\pi/x} \exp(-4\pi^2/x)$.

Finally, exponential tilting of the endpoint law leaves the binary coordinates independent and turns them into truncated exponentials. This yields a phase-resolved mod-Gamma limit, a central limit theorem with Berry–Esseen error $\mathcal{O}((\log s)^{-1/2})$, and a rigorous Lambert- W_{-1} localization law. A universal endpoint-transfer theorem then converts the exact negative-Laplace expansion into all-orders asymptotics for ordinary moments and centered even moments. In particular it supplies the previously missing bridge from the exact q -binomial/multiple-zeta coefficients of the sinc determinant to the Lambert- W endpoint saddle.

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1 Corpus audit, baseline, and novelty discipline

1.1 Scope of the audit

The audited source tree is `Analysis/FabiusFunction/docs` at the commit printed on the title page. It contains fifty-seven standalone $\text{\LaTeX}$ manuscripts when historical revisions, source transcriptions of reference papers, and the live frontier drafts are counted separately. They fall into five overlapping families:

1. the current primary exposition, Lean walkthrough, glossary, and non-elementarity study;
2. archived primary expositions, missing-parts supplements, and q -calculus introductions;
3. exact-dyadic, Thue–Morse, repeated-integration, inverse-function, and small-argument studies;
4. the consolidated non-formalized frontier and its sequence of frontier reports;
5. the multi-wave Fourier-decay campaign and the transcribed reference literature.

The revisions were not treated as independent evidence: later corrections supersede earlier drafts. In particular, the current Fourier audit already contains the Thue–Morse lobe-sign theorem, exact zero orders, the renormalization identity, and the global sharp decay envelope. Those are baseline results here, not rediscoveries.

1.2 What was already present

The corpus supplies, among much else, the following ingredients used below:

- the random-series realization of the bounded Fabius distribution and of Rvachev’s even density;
- the sinc product, its integer zero multiplicities $a_m = 1 + v_2(m)$, and its dyadic renormalization;
- exact moment recurrences, rationality, finite convolution models, and dyadic values expressed through Gaussian binomial coefficients;
- spectral-zeta and determinant viewpoints for the Wick-rotated product;
- the exact negative-Laplace decomposition

$$\log L(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{\log s}{2} + R(\log_2 s) + E(s),$$

where R is one-periodic and $E(s)$ is positive and exponentially small;

- Lambert- W_{-1} descriptions of the small-argument endpoint saddle;
- exact Thue–Morse signs of the Fourier lobes and sharp real-axis decay.

The goal was therefore not to enlarge the collection by cosmetic reformulation, but to find implications which genuinely cross the established boundaries between q -arithmetic, entire-function theory, probability, and endpoint asymptotics.

1.3 Status ledger

ID	Result	Status
A	Gaussian q -binomial/occupancy expansion of every even moment in symmetrized multiple zeta values, with a partition-lattice reduction to ordinary even zeta values.	proved here

ID	Result	Status
B	Type-I Laguerre–Pólya classification, PF_∞ coefficients, hyperbolic shifted Jensen polynomials, and the resulting ultra-log-concave moment hierarchy.	proved here
C	Strict sub-Gaussianity with the exact variance proxy, together with non-infinite-divisibility.	proved here
D	Explicit generalized gamma-convolution dual, discrete Thorin measure, cumulant duality, and an exact $q = 1/4$ perpetuity.	proved here
E	Thue–Morse lobe sign as parity of cumulative Thorin mass.	proved here
F	Exact Jacobi-theta completion of the Lévy heat kernel, including its first exponentially flat correction.	proved here
G	Phase-resolved mod-Gamma convergence of the endpoint tilt and a Berry–Esseen central limit theorem.	proved here
H	Universal endpoint-transfer expansion, applied to ordinary moments, centered moments, and the large-order q/MZV coefficients.	proved here
I	Phase-resolved local limit and full Edgeworth expansion under the endpoint tilt.	frontier proposal

The word “new” in this report means new to the audited corpus after targeted searches for the exact structures used below. It is not a claim of priority over every article in the mathematical literature. The general theories of multiple zeta values, Pólya-frequency sequences, generalized gamma convolutions, Mellin harmonic sums, and Lambert W are classical; what is new here is their exact synthesis for the Fabius–Rvachev product and the resulting cross-representation theorems.

2 Normalization and the master determinant

2.1 The random series

Let $(V_k)_{k \geq 1}$ be independent uniform random variables on $[0, 1]$, and let $U_k = 2V_k - 1$, uniform on $[-1, 1]$. Define

$$X = \sum_{k \geq 1} 2^{-k} V_k, \quad Y = 2X - 1 = \sum_{k \geq 1} 2^{-k} U_k. \quad (1)$$

Then $X \in [0, 1]$, $Y \in [-1, 1]$, and

$$X \stackrel{d}{=} 1 - X, \quad Y \stackrel{d}{=} -Y. \quad (2)$$

The distribution function of X is the bounded Fabius function, and the density of Y is Rvachev’s up-function. We write

$$m_n = \mathbb{E}X^n, \quad \mu_{2n} = \mathbb{E}Y^{2n}. \quad (3)$$

2.2 Fourier and Wick-rotated products

With $\text{sinc } z = \sin z / z$ and the Fourier convention $\Phi(z) = \mathbb{E}e^{-2\pi i z Y}$,

$$\Phi(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{a_m}, \quad a_m = 1 + v_2(m). \quad (4)$$

The first product gives the exact renormalization

$$\Phi(z) = \text{sinc}(\pi z) \Phi(z/2), \quad (5)$$

whereas the second shows that the zero at a nonzero integer m has order a_m .

The central entire function of this report is the Wick rotation

$$\mathcal{A}(w) = \Phi(i\sqrt{w}) = \prod_{m \geq 1} \left(1 + \frac{w}{m^2}\right)^{a_m} = \prod_{j \geq 0} \prod_{n \geq 1} \left(1 + \frac{4^{-j}w}{n^2}\right). \quad (6)$$

It is independent of the branch of $\sqrt{w}$ because it is entire in w . Set

$$\mathcal{A}(w) = \sum_{r \geq 0} C_r w^r. \quad (7)$$

Since

$$\mathbb{E}e^{tY} = \mathcal{A}\left(\frac{t^2}{4\pi^2}\right), \quad (8)$$

we have the exact coefficient–moment correspondence

$$C_r = \frac{(2\pi)^{2r}}{(2r)!} \mu_{2r}. \quad (9)$$

2.3 The spectral power sums

The logarithm of [equation \(6\)](#) is

$$\log \mathcal{A}(w) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} p_r w^r, \quad p_r = \sum_{m \geq 1} \frac{a_m}{m^{2r}} = \frac{\zeta(2r)}{1 - 4^{-r}}. \quad (10)$$

Consequently

$$rC_r = \sum_{j=1}^r (-1)^{j+1} p_j C_{r-j}, \quad C_0 = 1. \quad (11)$$

This Newton recurrence is already enough to prove rationality of all μ_{2r} because every $\zeta(2j)$ is a rational multiple of π^{2j} . The next section gives a different, genuinely q -binomial route which keeps the dyadic layers visible.

3 A q -binomial and multiple-zeta formula for every even moment

Throughout this section

$$q = \frac{1}{4}. \quad (12)$$

3.1 Finite depth and one-site occupancy

For $J \geq 1$ truncate the dyadic layers:

$$\mathcal{A}_J(w) = \prod_{j=0}^{J-1} \prod_{n \geq 1} \left(1 + \frac{q^j w}{n^2}\right) = \sum_{r \geq 0} C_r^{(J)} w^r. \quad (13)$$

At a fixed spectral site n , the finite q -binomial theorem gives

$$\prod_{j=0}^{J-1} (1 + q^j x) = \sum_{r=0}^J b_{J,r} x^r, \quad b_{J,r} = q^{r(r-1)/2} \begin{bmatrix} J \\ r \end{bmatrix}_q. \quad (14)$$

Thus r records the number of dyadic layers occupying the same integer frequency n . In the infinite-depth limit,

$$b_r = \lim_{J \rightarrow \infty} b_{J,r} = \frac{q^{r(r-1)/2}}{(q; q)_r}, \quad (q; q)_r = \prod_{j=1}^r (1 - q^j). \quad (15)$$

For $q = 1/4$,

$$b_1 = \frac{4}{3}, \quad b_2 = \frac{16}{45}, \quad b_3 = \frac{64}{2835}. \quad (16)$$

3.2 Distinct-index zeta sums

Let $\lambda = (\lambda_1, \dots, \lambda_\ell)$ be a partition of r , and let $m_j(\lambda)$ denote the multiplicity of the part j . Define the symmetrized distinct-index zeta sum

$$Z_\lambda = \sum_{\substack{n_1, \dots, n_\ell \geq 1 \\ n_i \neq n_j \ (i \neq j)}} \frac{1}{n_1^{2\lambda_1} \dots n_\ell^{2\lambda_\ell}}. \quad (17)$$

This notation keeps the assignment of distinct spectral sites to the parts of λ explicit.

Theorem 3.1 (q -binomial/MZV moment formula). *For every $r \geq 0$ and $J \geq 1$,*

$$C_r^{(J)} = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} b_{J, \lambda_i}. \quad (18)$$

Passing to infinite depth gives

$$C_r = \sum_{\lambda \vdash r} \frac{Z_\lambda}{\prod_{j \geq 1} m_j(\lambda)!} \prod_{i=1}^{\ell(\lambda)} \frac{q^{\lambda_i(\lambda_i-1)/2}}{(q; q)_{\lambda_i}}. \quad (19)$$

Moreover

$$Z_\lambda = \sum_{\sigma \in S_\ell} \zeta(2\lambda_{\sigma(1)}, \dots, 2\lambda_{\sigma(\ell)}), \quad (20)$$

where $\zeta(s_1, \dots, s_\ell) = \sum_{n_1 > \dots > n_\ell \geq 1} n_1^{-s_1} \dots n_\ell^{-s_\ell}$.

Proof. Apply [equation \(14\)](#) at each spectral site n :

$$\mathcal{A}_J(w) = \prod_{n \geq 1} \left(\sum_{r=0}^J b_{J,r} \frac{w^r}{n^{2r}} \right).$$

A monomial of total degree r is specified by a finite set of distinct sites $n_1, \dots, n_\ell$ and positive occupancies $\lambda_1 + \dots + \lambda_\ell = r$. Sorting the occupancies gives a partition λ . The sum over ordered, pairwise distinct sites is Z_λ ; permuting sites carrying equal occupancies does not change the monomial, hence the divisor $\prod_j m_j(\lambda)!$. This proves [equation \(18\)](#). Since the products converge normally on compact subsets and $b_{J,r} \rightarrow b_r$, coefficientwise convergence proves [equation \(19\)](#).

Finally, every pairwise-distinct ℓ -tuple has a unique strict ordering. Summing over the $\ell!$ possible orderings of the labelled coordinates gives [equation \(20\)](#); repeated parts correctly produce repeated MZVs, matching the labelled sum in [equation \(17\)](#). $\square$

3.3 Reduction to ordinary zeta values

The MZV form is conceptually natural, but in the present fully symmetrized combination it collapses to ordinary zeta values by Möbius inversion on the partition lattice.

Theorem 3.2 (Partition-lattice reduction). *Let Π_ℓ be the lattice of set partitions of $\{1, \dots, \ell\}$. Then*

$$Z_\lambda = \sum_{\pi \in \Pi_\ell} \left(\prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)! \right) \prod_{B \in \pi} \zeta \left(2 \sum_{i \in B} \lambda_i \right). \quad (21)$$

Consequently [equation \(19\)](#) is a finite expression involving only q -Pochhammer factors and ordinary even zeta values.

Proof. Start with the unrestricted sum over $(n_1, \dots, n_\ell)$. Coincidences among indices are encoded by set partitions: a block B means that all n_i with $i \in B$ coincide. Möbius inversion from the full coincidence lattice to the discrete partition has Möbius factor $(-1)^{|B|-1} (|B| - 1)!$ on each block. The sum attached to a block is $\zeta(2 \sum_{i \in B} \lambda_i)$. Multiplication over blocks yields [equation \(21\)](#). $\square$

Corollary 3.3 (A second rationality proof). *Every centered even moment μ_{2r} is rational. More precisely, $C_r/\pi^{2r} \in \mathbb{Q}$, and [equation \(9\)](#) gives $\mu_{2r} \in \mathbb{Q}$.*

Remark 3.4. The usual Newton recurrence [equation \(11\)](#) forgets where repeated spectral integers came from. Formula [equation \(19\)](#) retains that information: the Gaussian q -binomial factor is an occupancy weight across dyadic layers, while the MZV records the strict ordering of distinct integer frequencies. This is the precise meeting point of the q -binomial and sinc representations.

3.4 First examples

Using either [equation \(19\)](#) or [equation \(11\)](#),

$$C_1 = \frac{2\pi^2}{9}, \quad \mu_2 = \frac{1}{9}, \quad (22)$$

$$C_2 = \frac{38\pi^4}{2025}, \quad \mu_4 = \frac{19}{675}, \quad (23)$$

$$C_3 = \frac{2332\pi^6}{2679075}, \quad \mu_6 = \frac{583}{59535}, \quad (24)$$

$$C_4 = \frac{265618\pi^8}{10247461875}, \quad \mu_8 = \frac{132809}{32531625}. \quad (25)$$

For example, at degree two the partitions (2) and (1, 1) give

$$C_2 = b_2 \zeta(4) + b_1^2 \zeta(2, 2). \quad (26)$$

The identity $2\zeta(2, 2) = \zeta(2)^2 - \zeta(4)$ immediately recovers the Newton form $(p_1^2 - p_2)/2$.

4 Laguerre–Pólya structure and an infinite hierarchy of moment inequalities

4.1 Classification of the Wick-rotated determinant

Theorem 4.1 (Type-I Laguerre–Pólya classification). *The entire function $\mathcal{A}$ belongs to the type-I Laguerre–Pólya class. It has genus zero, order $1/2$, and only real nonpositive zeros, namely $-m^2$ with multiplicity $a_m = 1 + v_2(m)$.*

Proof. The spectral sum

$$\sum_{m \geq 1} \frac{a_m}{m^2} = \frac{\zeta(2)}{1 - 4^{-1}} = \frac{2\pi^2}{9}$$

converges. Therefore the genus-zero product in [equation \(6\)](#) converges normally on compact sets. Every finite truncation is a polynomial with all zeros real and nonpositive; their locally uniform limit is $\mathcal{A}$, so $\mathcal{A}$ is type-I Laguerre–Pólya. The exponent of convergence of the zeros is $1/2$, because

$$\sum_{m \geq 1} \frac{a_m}{(m^2)^\rho} = \frac{\zeta(2\rho)}{1 - 4^{-\rho}}$$

converges exactly for $\rho > 1/2$. The support bound $|Y| \leq 1$ and [equation \(8\)](#) show that $\mathcal{A}(w)$ has growth $\exp(\mathcal{O}(\sqrt{|w|}))$, completing the order statement. $\square$

4.2 Pólya frequency and Jensen hyperbolicity

Write

$$\mathcal{A}(w) = \sum_{r \geq 0} \gamma_r \frac{w^r}{r!}, \quad \gamma_r = r! C_r = \frac{r! (2\pi)^{2r}}{(2r)!} \mu_{2r}. \quad (27)$$

Theorem 4.2 (PF $_{\infty}$ and Jensen polynomials). *The ordinary coefficient sequence $(C_r)_{r \geq 0}$ is a Pólya-frequency sequence of infinite order. Equivalently, every finite minor of the Toeplitz matrix*

$$T = (C_{j-i})_{i,j \geq 0}, \quad C_k = 0 \quad (k < 0), \quad (28)$$

is nonnegative.

Furthermore, for every $n, d \geq 0$, the shifted Jensen polynomial

$$J_n^{(d)}(x) = \sum_{j=0}^d \binom{d}{j} \gamma_{n+j} x^j \quad (29)$$

has only real nonpositive zeros.

Proof. The Aissen–Schoenberg–Whitney characterization says that the generating function of a PF $_{\infty}$ sequence has the form

$$Cz^m e^{\sigma z} \frac{\prod_i (1 + \alpha_i z)}{\prod_j (1 - \beta_j z)}, \quad \alpha_i, \beta_j, \sigma \geq 0, \quad \sum_i \alpha_i + \sum_j \beta_j < \infty.$$

Equation [equation \(6\)](#) is exactly this form with no denominator, no exponential factor, and $\alpha_i = 1/m^2$ repeated a_m times. This proves the PF $_{\infty}$ assertion. Jensen’s characterization of the Laguerre–Pólya class, applied to [equation \(27\)](#), gives hyperbolicity of all shifted Jensen polynomials. Their coefficients are positive, so their real zeros cannot be positive. $\square$

Corollary 4.3 (Ultra-log-concavity of even moments). *For every $r \geq 1$,*

$$\gamma_r^2 \geq \gamma_{r-1} \gamma_{r+1}. \quad (30)$$

Equivalently,

$$\boxed{\mu_{2r}^2 \geq \frac{2r-1}{2r+1} \mu_{2r-2} \mu_{2r+2}.} \quad (31)$$

Hence the sequence

$$\frac{2\pi^2}{2r+1} \frac{\mu_{2r+2}}{\mu_{2r}} = \frac{\gamma_{r+1}}{\gamma_r} \quad (32)$$

is nonincreasing.

Proof. The discriminant of $J_{r-1}^{(2)}(x) = \gamma_{r-1} + 2\gamma_r x + \gamma_{r+1} x^2$ is nonnegative. Substituting [equation \(27\)](#) and cancelling the common powers of 2π gives [equation \(31\)](#). The ratio statement is the usual consequence of log-concavity. $\square$

Remark 4.4 (Higher inequalities). The quadratic inequality is only the first shadow of [theorem 4.2](#). Every Toeplitz minor and every discriminant or subdiscriminant of every $J_n^{(d)}$ gives a polynomial inequality among exact Fabius moments. This hierarchy is algorithmic and strictly stronger than checking Hankel positivity, which follows for unrelated reasons from the fact that (μ_{2r}) is the moment sequence of Y^2 .

5 Optimal sub-Gaussian concentration without infinite divisibility

Theorem 5.1 (Strict optimal sub-Gaussianity). *For every real t ,*

$$\mathbb{E}e^{tY} \leq \exp\left(\frac{t^2}{18}\right), \quad (33)$$

with strict inequality for $t \neq 0$. The variance proxy $1/9$ is optimal. Consequently, for $u \geq 0$,

$$\mathbb{P}(|Y| \geq u) \leq 2e^{-9u^2/2}, \quad \mathbb{P}\left(\left|X - \frac{1}{2}\right| \geq u\right) \leq 2e^{-18u^2}. \quad (34)$$

Proof. From [equations \(6\) and \(8\)](#),

$$\log \mathbb{E}e^{tY} = \sum_{m \geq 1} a_m \log\left(1 + \frac{t^2}{4\pi^2 m^2}\right).$$

The strict inequality $\log(1+x) < x$ for $x > 0$ gives, when $t \neq 0$,

$$\log \mathbb{E}e^{tY} < \frac{t^2}{4\pi^2} \sum_{m \geq 1} \frac{a_m}{m^2} = \frac{t^2}{18}.$$

At the origin equality holds. Since $\text{Var}(Y) = \mu_2 = 1/9$, no smaller quadratic proxy can dominate the moment generating function near zero. Chernoff optimization gives the two tail bounds. $\square$

Theorem 5.2 (The Fabius–Rvachev law is not infinitely divisible). *Neither X nor Y is infinitely divisible; in particular, neither is self-decomposable.*

Proof. The characteristic function of Y is $t \mapsto \Phi(t/(2\pi))$, and [equation \(4\)](#) shows that it vanishes at $t = 2\pi m$ for every nonzero integer m . An infinitely divisible characteristic function is the exponential of a Lévy–Khintchine exponent and hence never vanishes. Translation and nonzero scaling preserve infinite divisibility, so the assertion transfers between X and Y . $\square$

There is no contradiction between [theorems 5.1 and 5.2](#). The law is as concentrated as its variance permits in the sub-Gaussian sense, but it does not sit inside a convolution semigroup. The next section identifies a canonical object on the other side of the sinc determinant which *is* infinitely divisible.

6 The generalized gamma-convolution dual

6.1 Construction and Thorin measure

For $\tau > 0$, let $G_{j,n}^{(\tau)}$ be independent gamma variables with shape τ and rate n^2 :

$$\mathbb{E}e^{-wG_{j,n}^{(\tau)}} = \left(1 + \frac{w}{n^2}\right)^{-\tau}. \quad (35)$$

Theorem 6.1 (Spectral gamma dual). *The nonnegative random series*

$$V_\tau = \sum_{j \geq 0} \sum_{n \geq 1} q^j G_{j,n}^{(\tau)}, \quad q = \frac{1}{4}, \quad (36)$$

converges almost surely and in L^1 , with

$$\mathbb{E}V_\tau = \frac{2\tau\pi^2}{9}. \quad (37)$$

Its Laplace transform is exactly

$$\boxed{\mathbb{E}e^{-wV_\tau} = \mathcal{A}(w)^{-\tau}}, \quad w \geq 0. \quad (38)$$

Equivalently,

$$V_\tau \stackrel{d}{=} \sum_{m \geq 1} H_m, \quad H_m \sim \text{Gamma}(\tau a_m, \text{rate } m^2) \quad (39)$$

with independent summands. Thus V_τ is a generalized gamma convolution with discrete Thorin measure

$$\boxed{U_\tau = \tau \sum_{m \geq 1} a_m \delta_{m^2}.} \quad (40)$$

Proof. The expectation of the nonnegative series is

$$\tau \sum_{j,n} \frac{q^j}{n^2} = \frac{\tau \zeta(2)}{1-q} = \frac{2\tau\pi^2}{9} < \infty,$$

so the series is finite almost surely and converges in L^1 . Independence and [equation \(35\)](#) give

$$\mathbb{E}e^{-wV_\tau} = \prod_{j,n} \left(1 + \frac{q^j w}{n^2}\right)^{-\tau} = \mathcal{A}(w)^{-\tau}.$$

Regrouping equal rates uses the identity that the sum of independent gamma variables with a common rate is gamma with the sum of their shapes. The multiplicity of rate m^2 is a_m . Finally, the defining Laplace exponent of a generalized gamma convolution is $\int \log(1 + w/u) U(\mathrm{d}u)$; [equation \(40\)](#) reproduces $\tau \log \mathcal{A}(w)$. $\square$

Corollary 6.2 (Exact cumulants and cumulant duality). *For every $r \geq 1$,*

$$\boxed{\kappa_r(V_\tau) = \tau(r-1)! \frac{\zeta(2r)}{1-4^{-r}}.} \quad (41)$$

The even cumulants of Y are obtained from those of V_1 by

$$\boxed{\kappa_{2r}(Y) = (-1)^{r+1} \frac{(2r)!}{r!(2\pi)^{2r}} \kappa_r(V_1).} \quad (42)$$

Proof. A gamma variable with shape a and rate λ has r th cumulant $a(r-1)!/\lambda^r$. Summing [equation \(39\)](#) gives [equation \(41\)](#). Comparing the expansion

$$\log \mathbb{E}e^{tY} = \log \mathcal{A}(t^2/(4\pi^2))$$

with [equation \(10\)](#) proves [equation \(42\)](#). $\square$

6.2 An exact $q = 1/4$ perpetuity

Define the one-layer variable

$$B_\tau = \sum_{n \geq 1} G_{0,n}^{(\tau)}. \quad (43)$$

Euler's product for sinh gives

$$\mathbb{E}e^{-wB_\tau} = \left(\frac{\pi\sqrt{w}}{\sinh(\pi\sqrt{w})} \right)^\tau. \quad (44)$$

Corollary 6.3 (Gamma perpetuity). *Let V'_τ be an independent copy of V_τ . Then*

$$\boxed{V_\tau \stackrel{d}{=} B_\tau + \frac{1}{4}V'_\tau.} \quad (45)$$

This equation has a unique law on $[0, \infty)$ with finite first moment.

Proof. Split the layer $j = 0$ from equation (36); after shifting $j \mapsto j + 1$, the remaining sum is q times an independent copy. Uniqueness follows by iteration in the Wasserstein-1 metric, where the map $\nu \mapsto \text{Law}(B_\tau + qZ)$ is a contraction with factor q . $\square$

6.3 Thue–Morse parity as Thorin-mass parity

Let $s_2(N)$ be the number of 1's in the binary expansion of N . Legendre's identity $v_2(N!) = N - s_2(N)$ gives

$$\sum_{m \leq N} a_m = N + v_2(N!) = 2N - s_2(N). \quad (46)$$

Theorem 6.4 (Thue–Morse/Thorin parity law). *For every integer $N \geq 1$,*

$$\boxed{(-1)^{U_1([1, N^2])} = (-1)^{s_2(N)}.} \quad (47)$$

Hence, for every positive noninteger x ,

$$\boxed{\text{sgn } \Phi(x) = (-1)^{U_1([1, [x]^2])}.} \quad (48)$$

Proof. By equation (40), $U_1([1, N^2]) = \sum_{m \leq N} a_m$. Equation (46) has the same parity as $s_2(N)$. The current Fourier sign theorem gives $\text{sgn } \Phi(x) = (-1)^{s_2([x])}$ on each open lobe; substitution proves equation (48). $\square$

Remark 6.5. The Thue–Morse sign is therefore not merely an external combinatorial label. It is the parity of the spectral mass accumulated by the infinitely divisible dual before the corresponding squared zero. This interpretation links the real-axis sign changes of a non-infinitely-divisible compactly supported law to the Thorin measure of its reciprocal Wick rotation.

6.4 Complete Bernstein and Stieltjes consequences

Differentiating the logarithm gives

$$\frac{\mathcal{A}'(w)}{\mathcal{A}(w)} = \sum_{m \geq 1} \frac{a_m}{m^2 + w} = \int_{[1, \infty)} \frac{U_1(\mathrm{d}u)}{u + w}. \quad (49)$$

Thus $\mathcal{A}'/\mathcal{A}$ is a Stieltjes function, $\log \mathcal{A}$ is a complete Bernstein function, and $\mathcal{A}^{-\tau}$ is completely monotone for every $\tau > 0$. Standard Stieltjes theory then supplies convergent continued fractions and alternating Padé bounds on the positive real axis. These are certified rational bounds for the reciprocal Wick-rotated sinc product, complementary to the product truncations used elsewhere in the corpus.

7 The Lévy heat kernel and its exact theta completion

7.1 Lévy density and Mellin transform

The gamma dual has Lévy–Khintchine representation

$$-\log \mathbb{E} e^{-wV_\tau} = \tau \int_0^\infty (1 - e^{-wx}) K(x) \frac{dx}{x}, \quad (50)$$

where

$$K(x) = \sum_{m \geq 1} a_m e^{-m^2 x} = \sum_{j \geq 0} \sum_{n \geq 1} e^{-4^j n^2 x}. \quad (51)$$

Its Mellin transform is

$$\int_0^\infty K(x) x^{s-1} dx = \Gamma(s) \frac{\zeta(2s)}{1 - 4^{-s}}, \quad \Re s > \frac{1}{2}. \quad (52)$$

The poles at

$$\omega_k = \frac{\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \quad (53)$$

produce a periodic fluctuation in $\log_4(1/x)$.

Put $L_2 = \log 2$ and

$$C_K = \frac{\gamma}{4L_2} - \frac{1}{4} - \frac{\log(2\pi)}{2L_2} = -1.36756152041694261136 \dots, \quad (54)$$

$$P_K(u) = \frac{1}{2L_2} \sum_{k \neq 0} \Gamma(\omega_k) \zeta(2\omega_k) e^{2\pi i k u}. \quad (55)$$

The series is real-valued and absolutely, exponentially convergent. Define

$$M_K(x) = \sqrt{\frac{\pi}{x}} - \frac{\log(1/x)}{4L_2} + C_K + P_K\left(\log_4 \frac{1}{x}\right). \quad (56)$$

Mellin residue calculus gives $K(x) = M_K(x) + \mathcal{O}_N(x^N)$ for every $N > 0$. The next theorem identifies the flat remainder exactly.

7.2 Exact completion

Let

$$\vartheta_+(t) = \sum_{n \geq 1} e^{-n^2 t}. \quad (57)$$

Theorem 7.1 (Exact Jacobi-theta completion). *For every $x > 0$,*

$$K(x) = M_K(x) - 2\sqrt{\frac{\pi}{x}} \sum_{r=0}^\infty 2^r \vartheta_+\left(\frac{4^{r+1}\pi^2}{x}\right). \quad (58)$$

In particular, as $x \downarrow 0$,

$$K(x) - M_K(x) = -2\sqrt{\frac{\pi}{x}} e^{-4\pi^2/x} \left(1 + \mathcal{O}(e^{-12\pi^2/x})\right). \quad (59)$$

Proof. From the layered definition,

$$K(x) - K(4x) = \vartheta_+(x). \quad (60)$$

The Jacobi transformation is

$$\vartheta_+(x) = \frac{1}{2} \sqrt{\frac{\pi}{x}} - \frac{1}{2} + \sqrt{\frac{\pi}{x}} \vartheta_+\left(\frac{\pi^2}{x}\right). \quad (61)$$

Periodicity of P_K and the elementary terms in [equation \(56\)](#) give

$$M_K(x) - M_K(4x) = \frac{1}{2} \sqrt{\frac{\pi}{x}} - \frac{1}{2}. \quad (62)$$

Thus $D(x) = K(x) - M_K(x)$ satisfies

$$D(x) - D(4x) = \sqrt{\frac{\pi}{x}} \vartheta_+\left(\frac{\pi^2}{x}\right). \quad (63)$$

Replace x by $x/4$, solve for $D(x)$, and iterate toward zero. The Mellin flatness $D(t) = \mathcal{O}_N(t^N)$ makes the terminal term vanish and yields

$$D(x) = -\sqrt{\frac{\pi}{x}} \sum_{r=1}^{\infty} 2^r \vartheta_+\left(\frac{4^r \pi^2}{x}\right),$$

which is [equation \(58\)](#). The first term is $(r, n) = (1, 1)$. The next possible exponent is $16\pi^2/x$, twelve units of π^2/x beyond the leading $4\pi^2/x$, proving [equation \(59\)](#). $\square$

Remark 7.2 (Size of the periodic ripple). The first nonzero Fourier coefficient of P_K is

$$c_1 = \frac{\Gamma(\pi i/L_2) \zeta(2\pi i/L_2)}{2L_2}, \quad |c_1| = 0.0011153475185173782067 \dots$$

Thus the first-harmonic peak-to-center amplitude is $2|c_1| = 0.0022306950370347564135 \dots$. The ripple is small but not formal noise; it is an exact complex-dimension contribution.

Table 2: The exact flat correction $K(x) - M_K(x)$ and the theta completion.

x	direct difference	theta-series value
1	$-2.5371492292 \times 10^{-17}$	$-2.5371492292 \times 10^{-17}$
2	$-6.7059525212 \times 10^{-9}$	$-6.7059525212 \times 10^{-9}$
5	$-5.9029369697 \times 10^{-4}$	$-5.9029369697 \times 10^{-4}$

8 Endpoint tilting: truncated exponentials, mod-Gamma limits, and Lambert W

8.1 Exact negative-Laplace decomposition

Let

$$L(s) = \mathbb{E}e^{-sX}, \quad g(s) = \log L(s), \quad s > 0. \quad (64)$$

The binary random series gives

$$L(s) = \prod_{k \geq 1} \frac{1 - e^{-s/2^k}}{s/2^k}. \quad (65)$$

The current exact endpoint theory supplies a real-analytic one-periodic function R and an exponentially small positive error E such that

$$g(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{\log s}{2} + R(\log_2 s) + E(s), \quad (66)$$

with

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (67)$$

Termwise differentiation of the defining forward-tail series gives the corresponding exponentially small bounds for each fixed derivative of E .

8.2 The tilted binary coordinates

Define the exponentially tilted probability measure

$$\mathbb{P}_s(dx) = \frac{e^{-sx}}{L(s)} \mathbb{P}(X \in dx). \quad (68)$$

Let

$$\lambda_k = s2^{-k}, \quad W_{k,s} = \lambda_k V_k. \quad (69)$$

Theorem 8.1 (Exact tilted-coordinate law). *Under $\mathbb{P}_s$, the coordinates V_k remain independent. The variables $W_{k,s}$ have densities*

$$f_{\lambda_k}(w) = \frac{e^{-w}}{1 - e^{-\lambda_k}} \mathbf{1}_{(0, \lambda_k)}(w), \quad (70)$$

and

$$sX = \sum_{k \geq 1} W_{k,s}. \quad (71)$$

Proof. The Radon–Nikodym factor factorizes:

$$e^{-sX} = \prod_{k \geq 1} e^{-\lambda_k V_k}.$$

Normalizing each uniform coordinate gives density $\lambda_k e^{-\lambda_k v} / (1 - e^{-\lambda_k})$ on $0 < v < 1$. The change of variable $w = \lambda_k v$ yields [equation \(70\)](#), and summing gives [equation \(71\)](#). $\square$

This theorem is the probabilistic mechanism behind the endpoint saddle. About $\log_2 s$ coordinates have $\lambda_k \gtrsim 1$ and behave almost as independent unit exponentials; the remaining coordinates form a uniformly bounded tail.

8.3 Tilted cumulants and exact phase dependence

Let

$$K_r(s) = \kappa_r^{(s)}(sX) = (-1)^r s^r g^{(r)}(s), \quad (72)$$

where the superscript indicates cumulants under $\mathbb{P}_s$. Put

$$\alpha(s) = K_1(s) = -sg'(s), \quad \beta(s) = K_2(s) = s^2 g''(s). \quad (73)$$

Writing $\theta = \log_2 s$ and differentiating [equation \(66\)](#) gives

$$\alpha(s) = \log_2 s - \frac{1}{2} - \frac{R'(\theta)}{\log 2} + \mathcal{O}(se^{-s}), \quad (74)$$

$$\beta(s) = \alpha(s) - \frac{1}{\log 2} + \frac{R''(\theta)}{(\log 2)^2} + \mathcal{O}(s^2 e^{-s}). \quad (75)$$

In particular $\beta(s) = \log_2 s + \mathcal{O}(1)$. The cumulants obey the exact recursion

$$K_{r+1}(s) = rK_r(s) - sK'_r(s). \quad (76)$$

Theorem 8.2 (Phase-resolved mod-Gamma convergence). *Let $s_j \rightarrow \infty$ with fractional parts $\{\log_2 s_j\} \rightarrow \theta \in [0, 1]$. For real $t < 1$, locally uniformly in t ,*

$$\mathbb{E}_{s_j} e^{ts_j X} (1-t)^{\log_2 s_j} \longrightarrow \Psi_\theta(t), \quad (77)$$

where

$$\Psi_\theta(t) = \exp \left[-\frac{\log^2(1-t)}{2 \log 2} + \frac{1}{2} \log(1-t) + R(\theta + \log_2(1-t)) - R(\theta) \right]. \quad (78)$$

The convergence also holds in a complex neighborhood of the origin.

Proof. The tilted moment generating function is exact:

$$\mathbb{E}_s e^{tsX} = \frac{L((1-t)s)}{L(s)}. \quad (79)$$

Insert [equation \(66\)](#). If $v = \log(1-t)$ and $u = \log s$, the quadratic part contributes

$$-\frac{(u+v)^2 - u^2}{2 \log 2} + \frac{v}{2} = -\frac{uv}{\log 2} - \frac{v^2}{2 \log 2} + \frac{v}{2}.$$

Multiplication by $(1-t)^{\log_2 s}$ cancels the term $-uv/\log 2$. Periodicity and the assumed phase convergence give the R -difference in [equation \(78\)](#); the two exponentially small errors vanish locally uniformly. The same argument works for complex t near zero using the principal logarithm. $\square$

Corollary 8.3 (Phase-resolved cumulants). *For every fixed $r \geq 1$,*

$$K_r(s) = (r-1)! \log_2 s + \eta_r(\{\log_2 s\}) + o(1), \quad (80)$$

where η_r is an explicit analytic one-periodic function obtained by r differentiations of $\log \Psi_\theta(t)$ at $t = 0$.

8.4 Central and local concentration under the tilt

Theorem 8.4 (Tilted central limit theorem with Berry–Esseen rate). *As $s \rightarrow \infty$,*

$$\frac{sX - \alpha(s)}{\sqrt{\beta(s)}} \Longrightarrow N(0, 1). \quad (81)$$

Moreover

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_s \left(\frac{sX - \alpha(s)}{\sqrt{\beta(s)}} \leq x \right) - \Phi_N(x) \right| = \mathcal{O}((\log s)^{-1/2}), \quad (82)$$

where Φ_N is the standard normal distribution function.

Proof. Use the independent triangular array in [theorem 8.1](#). If $\lambda \geq 1$, a truncated exponential has every fixed centered absolute moment bounded uniformly in λ . There are $\log_2 s + \mathcal{O}(1)$ such coordinates. If $\lambda < 1$, the corresponding variable lies in an interval of length λ , and the sum of its third centered absolute moments is bounded by a convergent geometric series. Therefore

$$\sum_k \mathbb{E}_s |W_{k,s} - \mathbb{E}_s W_{k,s}|^3 = \mathcal{O}(\log s). \quad (83)$$

By [equation \(75\)](#), the total variance is asymptotic to $\log_2 s$. The Berry–Esseen inequality gives $\mathcal{O}(\log s)/(\log s)^{3/2}$, proving [equation \(82\)](#); the CLT follows. $\square$

8.5 Lambert- W_{-1} localization

The typical endpoint location under the tilt is

$$x_s = \mathbb{E}_s X = \frac{\alpha(s)}{s}. \quad (84)$$

Set

$$y = (\log 2) s x_s = (\log 2) \alpha(s). \quad (85)$$

Theorem 8.5 (Phase-corrected Lambert localization). *As $s \rightarrow \infty$,*

$$y e^{-y} = (\log 2) x_s \exp\left(\frac{\log 2}{2} + R'(\log_2 s) + o(1)\right). \quad (86)$$

Equivalently,

$$y = -W_{-1}\left(-(\log 2) x_s \exp\left(\frac{\log 2}{2} + R'(\log_2 s) + o(1)\right)\right). \quad (87)$$

After suppressing the bounded phase correction, one obtains the useful phase-blind equivalent

$$s \sim -\frac{W_{-1}(-\sqrt{2}(\log 2) x_s)}{(\log 2) x_s}. \quad (88)$$

Proof. Equation [equation \(74\)](#) gives

$$y = \log s - \frac{\log 2}{2} - R'(\log_2 s) + o(1).$$

Since $x_s = (y/\log 2)/s$, elimination of s yields [equation \(86\)](#). The relevant solution has $y \rightarrow \infty$, so it is the W_{-1} branch. A bounded multiplicative change in the small argument of W_{-1} changes its value by $\mathcal{O}(1)$; division by $y \asymp \log(1/x_s)$ proves the phase-blind equivalent. $\square$

Remark 8.6. Lambert W is not merely a formal inversion of the leading endpoint asymptotic. The tilted law concentrates in a Gaussian window of width $\sqrt{\beta(s)}/s \asymp \sqrt{\log s}/s$ around x_s , and [theorem 8.5](#) identifies that actual probabilistic center.

9 A universal endpoint-transfer theorem

9.1 Universal correction polynomials

For fixed $c > 0$ and an integer $N \geq 1$, define the one-sided endpoint moment

$$M_N(c) = \mathbb{E} \left[(1 - cX)^N \mathbf{1}_{\{X \leq 1/c\}} \right]. \quad (89)$$

Set $s = cN$ and $Z = sX$ under the tilted law $\mathbb{P}_s$. Then

$$\frac{M_N(c)}{L(cN)} = \mathbb{E}_{cN} \left[e^Z \left(1 - \frac{Z}{N}\right)^N \mathbf{1}_{\{Z \leq N\}} \right]. \quad (90)$$

Define universal polynomials P_r by the formal identity

$$e^z (1 - zt)^{1/t} = \exp\left(-\sum_{j \geq 2} \frac{z^j t^{j-1}}{j}\right) = \sum_{r \geq 0} P_r(z) t^r. \quad (91)$$

The first four are

$$\begin{aligned} P_0(z) &= 1, \\ P_1(z) &= -\frac{z^2}{2}, \\ P_2(z) &= \frac{z^4}{8} - \frac{z^3}{3}, \\ P_3(z) &= -\frac{z^6}{48} + \frac{z^5}{6} - \frac{z^4}{4}. \end{aligned} \tag{92}$$

Theorem 9.1 (Universal endpoint transfer). *Fix $c > 0$ and an integer $R \geq 0$. As $N \rightarrow \infty$,*

$$\frac{M_N(c)}{L(cN)} = \sum_{r=0}^R \frac{\mathbb{E}_{cN} P_r(Z)}{N^r} + \mathcal{O}_{R,c} \left(\frac{(\log N)^{2R+2}}{N^{R+1}} \right), \quad Z = cNX. \tag{93}$$

Every coefficient is an explicit polynomial in the tilted cumulants $K_1(cN), \dots, K_{2r}(cN)$ and hence in the periodic endpoint function R and its derivatives.

Proof. Equation [equation \(90\)](#) is obtained by multiplying and dividing the integrand by e^{-cNX} . On $0 \leq Z \leq (\log N)^2$, Taylor’s theorem applied to [equation \(91\)](#) gives a remainder bounded by

$$C_{R,c} N^{-R-1} (1 + Z)^{2R+2}.$$

The mod-Gamma estimate, or directly the independent truncated-exponential representation, implies

$$\mathbb{E}_s Z^k = \mathcal{O}_k((\log s)^k) \tag{94}$$

for each fixed k . This gives the displayed remainder on the central region.

For the complement, use Markov’s inequality with any fixed $0 < t < 1$ and [equation \(79\)](#). The mod-Gamma theorem gives $\mathbb{E}_s e^{tZ} = \mathcal{O}_t(s^{-\log_2(1-t)})$, only a fixed power of s , while $\exp(-t(\log N)^2)$ beats every power of N . The tail, including the region $Z > N$, is therefore smaller than the asserted error. Finally, raw moments are complete Bell polynomials in cumulants, which proves the last assertion. $\square$

Lean status. The finite algebraic last sentence is now formal. Under an arbitrary probability measure, with integrability assumed only for the powers occurring in a chosen polynomial, the theorem `integral_eval2_eq_sum_completeBell_momentCumulant` in `PolynomialExpectationCumulant.lean` rewrites its expectation as the coefficient-weighted complete Bell transform of the formal cumulants of its raw moments. The universal polynomials P_r , their formal exponential identity, recurrence, scalar evaluation, and first values are developed in `EndpointTransferPolynomials.lean`. These finite identities do *not* identify the formal cumulants with the analytic tilted cumulants $K_r(s)$, nor do they formalize the asymptotic expansion [\(93\)](#), the mod-Gamma estimate, or its remainder and tail bounds; those analytic steps remain frontier work.

9.2 The first two corrections

Write, at $s = cN$,

$$a = K_1(s) = \alpha(s), \quad b = K_2(s) = \beta(s), \quad c_3 = K_3(s), \quad c_4 = K_4(s). \tag{95}$$

Then

$$\mathbb{E}_s Z^2 = a^2 + b, \tag{96}$$

$$\mathbb{E}_s Z^3 = a^3 + 3ab + c_3, \tag{97}$$

$$\mathbb{E}_s Z^4 = a^4 + 6a^2b + 3b^2 + 4ac_3 + c_4. \tag{98}$$

Thus

$$\frac{M_N(c)}{L(cN)} = 1 - \frac{\mathbb{E}_s Z^2}{2N} + \frac{\mathbb{E}_s Z^4/8 - \mathbb{E}_s Z^3/3}{N^2} + \mathcal{O}\left(\frac{(\log N)^6}{N^3}\right). \quad (99)$$

9.3 Ordinary Fabius moments

By symmetry $X \stackrel{d}{=} 1 - X$,

$$m_n = \mathbb{E}X^n = \mathbb{E}(1 - X)^n = M_n(1). \quad (100)$$

Corollary 9.2 (Large ordinary moments). *With all tilted quantities evaluated at $s = n$,*

$$\boxed{\frac{m_n}{L(n)} = 1 - \frac{\alpha(n)^2 + \beta(n)}{2n} + \frac{\mathbb{E}_n Z^4/8 - \mathbb{E}_n Z^3/3}{n^2} + \mathcal{O}\left(\frac{(\log n)^6}{n^3}\right)}. \quad (101)$$

In logarithmic form,

$$\log m_n = g(n) - \frac{\alpha(n)^2 + \beta(n)}{2n} + \mathcal{O}\left(\frac{(\log n)^4}{n^2}\right). \quad (102)$$

If $u = \log n$, $\theta = u/\log 2$, and

$$A_n = \frac{u}{\log 2} - \frac{1}{2} - \frac{R'(\theta)}{\log 2}, \quad (103)$$

then

$$\begin{aligned} \log m_n = & -\frac{u^2}{2\log 2} + \frac{u}{2} + R(\theta) \\ & - \frac{1}{2n} \left[A_n^2 + A_n - \frac{1}{\log 2} + \frac{R''(\theta)}{(\log 2)^2} \right] + \mathcal{O}\left(\frac{u^4}{n^2}\right). \end{aligned} \quad (104)$$

9.4 Centered moments and the q /MZV coefficient bridge

Symmetry about $1/2$ gives

$$\mu_{2r} = \mathbb{E}(1 - 2X)^{2r} = 2M_{2r}(2). \quad (105)$$

Here $N = 2r$ and the Laplace scale is $s = cN = 4r$.

Corollary 9.3 (Large centered moments). *With the tilted quantities evaluated at $s = 4r$,*

$$\begin{aligned} \frac{\mu_{2r}}{2L(4r)} = & 1 - \frac{\alpha(4r)^2 + \beta(4r)}{4r} \\ & + \frac{\mathbb{E}_{4r} Z^4/8 - \mathbb{E}_{4r} Z^3/3}{4r^2} + \mathcal{O}\left(\frac{(\log r)^6}{r^3}\right). \end{aligned} \quad (106)$$

Consequently the exact coefficient C_r in the q -binomial/MZV expansion [equation \(19\)](#) satisfies

$$\boxed{\begin{aligned} \log C_r = & 2r \log(2\pi) - \log(2r)! + \log 2 + g(4r) \\ & - \frac{\alpha(4r)^2 + \beta(4r)}{4r} + \mathcal{O}\left(\frac{(\log r)^4}{r^2}\right). \end{aligned}} \quad (107)$$

Since R has period one, $R(\log_2(4r)) = R(\log_2 r)$: the same binary phase controls both the endpoint Laplace law and the large-order q /MZV coefficients.

Proof. Apply [theorem 9.1](#) to [equation \(105\)](#) and then use [equation \(9\)](#). Taking the logarithm changes the second-order remainder only by $\mathcal{O}((\log r)^4/r^2)$. $\square$

Equation [equation \(107\)](#) is the main bridge result. The left side is an exact coefficient assembled from Gaussian q -binomials and MZVs. The right side is governed by the endpoint Laplace product, its log-periodic correction, and the Lambert- W_{-1} localization of the tilted mass. Neither side is an approximate restatement of the other; the transfer theorem proves why they must agree at large order and supplies computable corrections.

9.5 Numerical validation

The ordinary moments can be generated exactly by

$$(2^n - 1)m_n = \sum_{k=1}^n \binom{n}{k} \frac{m_{n-k}}{k+1}, \quad m_0 = 1. \quad (108)$$

Table 3 compares the exact ratio $m_n/L(n)$ with the one- and two-correction truncations of [equation \(101\)](#).

Table 3: Endpoint transfer for ordinary moments.

n	exact ratio	one correction	two corrections
20	0.650564349158	0.575338248846	0.658507245134
50	0.740712148665	0.698394360314	0.745157800500
100	0.809797566277	0.787758642159	0.811530920395
200	0.868346118425	0.858160052080	0.868893882715
500	0.924530320090	0.921307792046	0.924627286470
1000	0.952442261825	0.951188127893	0.952465665513

For centered moments the natural comparison is $\mu_{2r}/(2L(4r))$.

Table 4: Endpoint transfer for centered even moments.

r	exact ratio	one correction	two corrections
10	0.516638138954	0.334240960496	0.558725000470
20	0.623451024335	0.521571821828	0.641626320479
50	0.755044449961	0.716320104160	0.759255426702
100	0.834124742928	0.817440783081	0.835296570356

The second correction is already visibly asymptotic rather than merely qualitative. Its slower start for centered moments is expected: their relevant Laplace scale is $4r$, but the polynomial exponent is only $2r$, so the first correction carries the larger factor $(\alpha(4r)^2 + \beta(4r))/(4r)$.

10 Synthesis: one determinant, two probability laws, three asymptotic regimes

The results can be summarized by the single determinant

$$\mathcal{A}(w) = \underbrace{\Phi(i\sqrt{w})}_{\text{sinc/zero geometry}} = \underbrace{\mathbb{E} \exp(2\pi\sqrt{w} Y)}_{\text{compact law}} = \underbrace{(\mathbb{E} e^{-wV_1})^{-1}}_{\text{gamma-convolution dual}}. \quad (109)$$

Each equality exposes a different rigidity.

1. The sinc product gives integer zeros, dyadic multiplicities, and the Thue–Morse sign of real Fourier lobes.
2. The compact-law MGF gives exact centered moments, optimal sub-Gaussian concentration, and endpoint tilting.
3. The reciprocal Laplace transform gives a GGC, Thorin measure, complete Bernstein structure, and the Lévy heat kernel.
4. Expanding by dyadic layers produces Gaussian q -binomial occupancies; expanding by distinct integer sites produces symmetrized MZVs.
5. Mellin analysis of the spectral heat kernel and of the negative-Laplace product produces log-periodic phases. Endpoint transfer proves that those phases survive into large moment order.

This synthesis also clarifies a striking duality. The compact Fabius–Rvachev law is strictly variance-optimal sub-Gaussian but not infinitely divisible. The reciprocal Wick rotation is the Laplace transform of an explicitly infinitely divisible GGC. Their cumulants are related by the alternating formula [equation \(42\)](#); their sign structure is related by Thorin mass parity; and their asymptotics are tied by the same dyadic complex dimensions.

11 Next frontier targets

11.1 Phase-resolved Edgeworth and local limits

The proof of [theorem 8.4](#) contains more information than the stated Berry–Esseen bound. The independent summands have uniformly controlled moments, the variance diverges logarithmically, and [corollary 8.3](#) gives every cumulant with an explicit periodic constant term. This strongly suggests the following sharpened statement.

Conjecture 11.1 (Phase-resolved local Edgeworth expansion). *Along every sequence $\{\log_2 s_j\} \rightarrow \theta$, the tilted density of $(s_j X - \alpha(s_j))/\sqrt{\beta(s_j)}$ admits a full local Edgeworth expansion in powers of $(\log s_j)^{-1/2}$, uniformly on windows $|x| \leq (\log s_j)^{1/6-\varepsilon}$. Every coefficient is a polynomial in x whose constants are analytic one-periodic functions of θ .*

A proof should combine the explicit characteristic function $L((1 - it/s)s)/L(s)$ with a triangular-array Cramér bound. The compact support of each digit and the presence of many near-exponential coordinates make the nonlattice estimate plausible, but it should be checked uniformly in the binary phase.

11.2 All-orders moment transfer

Theorem [9.1](#) is already all-orders in principle. Two concrete tasks remain:

1. generate P_r by Bell polynomials and express $\mathbb{E}_s P_r(Z)$ directly in $R, R', \dots, R^{(2r)}$;
2. combine the result with Stirling’s expansion in [equation \(107\)](#) to produce a closed transseries for C_r , including the periodic phase and exponentially small endpoint terms.

This would turn the q /MZV coefficient sequence into a precise asymptotic probe of the negative-Laplace periodic function.

11.3 Inverse recovery of the periodic endpoint function

Equation [equation \(107\)](#) can be read in reverse. Exact rational moments produce exact C_r ; after subtracting the explicit Stirling and quadratic-log terms, the residual samples $R(\log_2 r)$ with a controlled error. Since $\{\log_2 r\}$ is dense modulo one, this suggests a certified reconstruction scheme for R from exact arithmetic alone. The correction terms involving R' and R'' make the inverse problem nonlinear but also provide derivative information. A practical scheme would alternate Fourier reconstruction of R with the transfer correction until the residual falls below the exact moment error bound.

11.4 A q -deformed gamma dual

The corpus defines symmetric q -integer deformations of the sinc determinant. Whenever the deformed zeros remain positive after Wick rotation, the same argument as [theorem 6.1](#) constructs a discrete Thorin measure at the squared q -integers. The main questions are:

- whether the deformed determinant remains type-I Laguerre–Pólya for the full parameter range;
- whether its Thorin mass admits a deformed digit-parity interpretation;
- how the $q \uparrow 1$ and $q \downarrow 0$ limits interact with the perpetuity and heat-kernel complex dimensions.

These questions are exact and experimentally accessible; no conjectural analytic continuation is needed to begin.

12 Lean formalization roadmap

The new results split naturally into modules with very different library requirements.

Post-snapshot algebraic progress. The later modules `ExponentialPartition.lean`, `ExponentialBell.lean`, and `MomentCumulantAlgebra.lean` now formalize, over every commutative $\mathbb{Q}$ -algebra, the universal weighted-partition formula for an exponential coefficient and identify it with the recursive `SaddleExpansion.expCoeff` engine, then factorially normalize it into named complete-Bell and inverse moment–cumulant transforms with the classical Bell recurrence and normalized inverse laws. This supplies reusable algebra for the Bell-polynomial sentence in [theorem 9.1](#) and for the coefficient stage of the proposed `MomentTransfer.lean`. It does not establish the tilted-cumulant formulas, their probabilistic interpretation, the Taylor and tail estimates, or the hierarchy-specific endpoint identities. Consequently none of the status labels in this report changes.

12.1 Low-risk algebraic modules

1. `QBinomialMZVMoments.lean`: prove the finite-depth occupancy expansion over finite spectral cutoffs, then pass first in the spectral cutoff and then in depth. The partition-lattice identity can be stated initially as a finite inclusion–exclusion theorem, avoiding a general MZV library.
2. `SpectralGammaCumulants.lean`: formalize the algebraic identities for Laplace exponents and cumulants before constructing the countable probability space. The equality $\kappa_r = \tau(r-1)!\zeta(2r)/(1-4^{-r})$ is mostly existing Dirichlet series infrastructure.
3. `ThorinParity.lean`: combine the already formalized lobe-sign theorem with Legendre’s digit-sum formula and a finite cumulative mass definition. This should be short.

12.2 Analytic modules

1. `GammaConvolutionDual.lean`: construct partial sums of independent gamma variables, prove L^1 convergence from the expectation sum, and identify the Laplace transform by dominated convergence.
2. `LevyHeatThetaCompletion.lean`: establish $K(x) - K(4x) = \vartheta_+(x)$, import or prove the Jacobi theta transform, and telescope the flat correction. The exact identity [equation \(58\)](#) is more formalization-friendly than a contour shift because all sums are positive or exponentially dominated.
3. `EndpointTilt.lean`: prove coordinate factorization under the tilted finite product, pass to the infinite law, and derive the cumulant recursion.
4. `MomentTransfer.lean`: define P_r through formal power series, prove a finite-order Taylor bound on a logarithmic window, and control the complement with the exact Laplace ratio.

12.3 Library-heavy modules

The Laguerre–Pólya and PF_∞ statements may require new general infrastructure. A pragmatic first formal target is finite truncation:

- prove real nonpositive roots of each finite polynomial;
- prove total positivity of its coefficient Toeplitz matrix by repeated convolution with $(1, \alpha)$;
- derive every fixed finite minor inequality for the limit by coefficient convergence.

This avoids formalizing the full Aissen–Schoenberg–Whitney theorem while still obtaining any requested moment inequality.

A Exact coefficients and normalized Turán ratios

For reference, the first five nontrivial centered moments are

r	μ_{2r}	
1	$1/9$	
2	$19/675$	
3	$583/59535$	
4	$132809/32531625$	
5	$46840699/24405225075$	(110)

The normalized Turán ratio

$$T_r = \frac{2r+1}{2r-1} \frac{\mu_{2r}^2}{\mu_{2r-2}\mu_{2r+2}} = \frac{\gamma_r^2}{\gamma_{r-1}\gamma_{r+1}} \quad (111)$$

is at least one. Its first values are

$$T_1 = \frac{25}{19} = 1.315789\dots, \quad T_2 = \frac{17689}{14575} = 1.21365\dots, \quad T_3 = 1.16829\dots \quad (112)$$

The approach toward one is consistent with the large-order endpoint asymptotic: the Jensen coefficients become asymptotically close to a geometric sequence on the scale of one index, while retaining strict log-concavity.

B Proof-dependency map

Result	Minimal dependencies
q /MZV formula	finite q -binomial theorem; normal convergence; set-partition Möbius inversion
Laguerre–Pólya	canonical product convergence; finite real-rooted truncations
PF_∞	Aissen–Schoenberg–Whitney characterization
Jensen hierarchy	Jensen characterization of the Laguerre–Pólya class
Sub-Gaussianity	$\log(1+x) < x$; exact spectral sum $p_1 = 2\pi^2/9$
Gamma dual	gamma Laplace transform; Tonelli; regrouping equal rates
Thorin parity	$\sum_{m \leq N} (1 + v_2(m)) = 2N - s_2(N)$; existing lobe sign
Theta completion	dyadic heat equation; Jacobi theta transform; telescoping
Mod-Gamma limit	exact negative-Laplace decomposition; periodicity of R
Berry–Esseen	tilted independent digits; third-moment bound; variance asymptotic
Moment transfer	exponential tilt; universal Taylor polynomials; tilted moment and tail bounds

C Corpus map used in the audit

The audit followed the repository’s revision structure rather than reading the directory as a flat bag of papers. The decisive source groups were:

Current expository core.

Fabius_Function_and_Rvachev_Up, the current Lean walkthrough, the mathematical glossary, and the non-elementarity article.

Archived foundational studies.

The earlier primary expositions, both missing-parts supplements, the two q -special-function expositions, the small-argument and inverse-function studies, and the early notes on Fabius asymptotics and iterated Thue–Morse summation.

Exact arithmetic and convergence studies.

The dyadic formula/alternative-representation papers, the two dyadic-to-asymptotic bridges, the Thue–Morse formula atlases, both Thue–Morse convergence-rate reports, and both repeated-integration reports.

Live frontier synthesis.

The consolidated non-formalized-research-frontiers manuscript, the two Fabius–Rvachev frontier reports, and the spectral-determinant/ q -deformation results bundle.

Fourier-decay campaign.

The original sinc-decay question, all numbered response waves, the final syntheses, gentle guide, comparative audit, and second-wave audit. The latest second-wave audit was treated as authoritative where older agents disagreed.

Reference transcriptions.

The corrected and source $\text{\LaTeX}$ transcriptions under docs/papers, including the work on the Fabius function, arithmetic properties, and lacunary products. They were used for historical and literature context, not as a substitute for checking the repository’s current normalization.

D Bibliographic notes

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